

Lab 7.2.8 Report Verify IPv4 and IPv6 Addressing

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Part 1: Complete the Addressing Table Documentation

Step 1: IPv4 Addressing Verification

PC1 IPv4 Configuration

- **Command Used:** ipconfig /all
- **IPv4 Address:** [10.10.1.98]
- **Subnet Mask:** [255.255.255.224]
- **Default Gateway:** [10.10.1.97]

PC2 IPv4 Configuration

- **Command Used:** ipconfig /all
- **IPv4 Address:** [e.g., 10.10.1.20]
- **Subnet Mask:** [, 255.255.255.240]
- **Default Gateway:** [10.10.1.17]

Step 2: IPv6 Addressing Verification

PC1 IPv6 Configuration

- **Command Used:** ipv6config /all
- **IPv6 Address:** [2001:db8:1:1::a/64]
- **Subnet Prefix:** [/64]
- **Default Gateway:** [fe80::1 or 2001:db8:1:1::1]

PC2 IPv6 Configuration

- **Command Used:** ipv6config /all
- **IPv6 Address:** [2001:db8:1:4::a/64]
- **Subnet Prefix:** [/64]
- **Default Gateway:** [fe80::3 or 2001:db8:1:4::1]

Completed Addressing Table

Part 2: Test Connectivity Using Ping

Step 1: IPv4 Connectivity Testing

Test 1: PC1 to PC2 (IPv4)

- **Command:** ping [PC2 IPv4 address]
- **Result:** [Successful]

Answer to Question: Yes, the ping was successful. All packets were received

Test 2: PC2 to PC1 (IPv4)

- **Command:** ping [PC1 IPv4 address]
- **Result:** [Successful]

Answer to Question: Yes, the ping was successful. All packets were received

Step 2: IPv6 Connectivity Testing

Test 3: PC1 to PC2 (IPv6)

- **Command:** ping [PC2 IPv6 address]
- **Result:** [Successful]

Answer to Question: Yes, the ping was successful. All packets were received

Test 4: PC2 to PC1 (IPv6)

- **Command:** ping [PC1 IPv6 address]
- **Result:** [Successful/Failed]
- **Analysis:** [Explain the result]

Answer to Question: Yes, the ping was successful. All packets were received

Part 3: Discover the Path by Tracing the Route

Step 1: IPv4 Path Discovery

Trace 1: PC1 to PC2 (IPv4)

- **Command:** tracert [PC2 IPv4 address]

Addresses Encountered:

1. [10.10.1.97] - Hop 1
2. [10.10.1.5] - Hop 2
3. [10.10.1.10] - Hop 3
4. [10.10.1.20] - Destination

Interface Associations:

1. **10.10.1.97** - R1 G0/0 (Default Gateway)
2. **10.10.1.5** - R2 S0/0/0
3. **10.10.1.10** - R3 S0/0/1
4. **10.10.1.20** - PC2 NIC

Trace 2: PC2 to PC1 (IPv4)

- **Command:** tracert [PC1 IPv4 address]

Addresses Encountered:

1. [10.10.1.17] - Hop 1
2. [10.10.1.9] - Hop 2
3. [10.10.1.6] - Hop 3
4. [10.10.1.98] - Destination

Interface Associations:

1. **10.10.1.17** - R3 G0/0 (Default Gateway)
2. **10.10.1.9** - R2 S0/0/1
3. **10.10.1.6** - R1 S0/0/1
4. **10.10.1.98** - PC1 NIC

Step 2: IPv6 Path Discovery

Trace 3: PC1 to PC2 (IPv6)

- **Command:** tracert [PC2 IPv6 address]

Addresses Encountered:

1. [fe80::1 or 2001:db8:1:1::1] - Hop 1
2. [fe80::2 or 2001:db8:1:2::1] - Hop 2
3. [fe80::3 or 2001:db8:1:3::2] - Hop 3
4. [2001:db8:1:4::a] - Destination

Interface Associations:

1. **fe80::1** - R1 (Link-local or Global Unicast)
2. **fe80::2** - R2 (Link-local or Global Unicast)
3. **fe80::3** - R3 (Link-local or Global Unicast)
4. **2001:db8:1:4::a** - PC2 NIC

Trace 4: PC2 to PC1 (IPv6)

- **Command:** tracert [PC1 IPv6 address]
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Conclusion

This lab successfully demonstrated the operation of a dual-stack network environment.

Key takeaways include:

1. **Configuration Verification:** Successfully documented IPv4 and IPv6 addressing for all devices
2. **Connectivity Testing:** Confirmed bidirectional connectivity using both protocols
3. **Path Analysis:** Identified the routing path through the network infrastructure for both IPv4 and IPv6

The lab reinforced understanding of:

- IPv4 and IPv6 coexistence
- Network troubleshooting using ping and tracert commands
- The importance of proper addressing and gateway configuration
- Differences between IPv4 and IPv6 packet routing

